

Bide Huang

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SUMMARY

Over 10 years industrial experience in software and system engineering, algorithm research and development in areas of robotics, automated driving, cloud application development and real time embedded system.

EXPERIENCE

Dek Technologies

Senior Software Engineer

Melbourne, Australia

January 2023 – Present

Reporting to Aftermarket Seeing Machine Engineering Manager, working with cross-discipline Engineering teams to deliver commercial driver monitoring application for Caterpillar.

- Achieved Milestone 1 customer deliverable through defining epic on smart camera diagnostics topic via FCP (Fovio communication protocol), developing stakeholder, system and software requirement, system architecture, scoping out design tasks for junior engineers and supporting test engineer for system integration test.

SwarmFarm Robotics

Senior Software Engineer

Melbourne, Australia

Jun 2021 – January 2023

Reporting to CPO, end to end responsibility for research and development of robot route planning software stack in C++; Lead a team of 3 developers to develop and commercialize a cloud based planning website - Farm Manager.

- Research and develop a generic route planning framework in C++ to generate traversal path for swarm robots to navigate in broadacre, turf and Orchard environment. Key algorithms are graph based A*, Hybrid A*, numerical optimization based path smoothing, non-holonomic Dubins path generation, and cost heuristics.
- Achieved path planning success rate to 98% and real time in robot traversal generation speed to 200ms by defining KPI, designing and deploying evaluation server to run large-scale farm dataset on cloud.
- Successfully scaled up production workflow for traversal graph model generation on cloud, by designing a serverless application to generate field models in parallel using AWS API gateway, Lambda, and S3.
- Significantly reduced manual effort (typically 20 man hours for a 100 hectares field) in robot optimal work plan creation by designing an algorithm to automatically generate path plan from surveyed geometries, e.g. tramlines, headlands, and obstacles.

Bosch

Lead Software & System Engineer

Melbourne, Australia

Jun 2018 – Jun 2021

Reporting to Engineering Manager, end to end responsibility for delivering software product/service for various customers including Victoria government and OEMs. Define engineering tasks as per project requirements, design system software architecture and develop feature, integrate algorithms, conduct testing and manage release, organise customer meetings and cross-regional collaborations.

- Designed and developed the prototype of electronic horizon for L2+ Advanced Driver Assist System, including horizon APIs for crowd-source map behaviour layer data extraction, ROS based horizon provider application, and RViz/DeckGL visualization tools.
- Successful demonstration of fully automated driving system for 2.5 million dollar government funded CAV Trial project. End-to-end responsibility of mapping and localization including GPS/Camera/Lidar/Radar sensors integration and calibration, data extraction and visualization, development of algorithm, and conduct mapping activities and road testing.
- Designed novel approach for developing HD planner map creation pipeline. Achieved centimetre accuracy of localization through multi-beam lidar intensity calibration, multi-sensor calibration, multi-modal localization performance optimization.

- Research and develop graph based Visual SLAM algorithm with stereo video camera for EKF sensor fusion; Creation and deployment of multi-layer spatial data visualization website for analysis and critical decision making.
- Successful delivery of ultrasonic sensor based waste assist function for JLR customer project. Designed and implemented embedded application, sensor data filtering algorithm, simulation and testing.
- Second runner-up of machine learning business innovation competition - Developed prototype of vision based lane marker detection using semantic segmentation for road quality analysis.

Blackmagic Design

Software Engineer

Melbourne, Australia

Jun 2014 – Jun 2018

Reporting to software engineering manager, responsible for real time embedded software development for high performance video post-production product, including drivers and application on micro-kernel based OS, API and multi-threaded cross platform video playback and capture applications and tools.

- Worked closely with other engineers to develop and release the world's smallest Ultra HD broadcast deck with professional 10-bit video/audio dual disk SD/UHS-II recording and playback.
- Lead engineer for successful delivery of Disney customer release for HyperDeck Mini Studio; Designed and implemented Advanced Media Protocol which enables video playback synchronization, frame-accurate timeline and slow motion control over serial and Ethernet.
- Lead engineer for Video Assist 4K product, successfully delivered and demonstrated 3D LUT video color correction preview feature and multi-language display and update in CES 2017.

Welling & Crossley

Electronics Engineer

Melbourne, Australia

Jan 2011 – Jun 2014

Reporting to chief R&D engineer, responsible for research and develop electronic schematics and PCBs, embedded software/device drivers for engine control unit, Qt based touch screen controller, variable speed pump controller, and remote controller for irrigation and power generator system.

- Outstanding employee achievement award for successful design and deployment of engine speed sensor for diesel and petrol power generators, which saves significant manufacturing cost.
- Successful demonstration of AS-2941 compliant touch screen fire pump controller to company CEO and stakeholders.
- Critical troubleshooting for half million dollar "Black Start" 500KVA power generator for Queensland water treatment plant.

QUALIFICATION

Coursera

Google Project Management: Professional Certificate

Online

Feb. 2023

Freiburg University

Robot Mapping

Online

Sep. 2018

Udacity

Self-Driving Car Nano Degree Certificate

Online

Nov. 2016 – Sep. 2017

Coursera

Machine Learning, Robotics Specialization

Online

June. 2015

RMIT University

Graduate Diploma in Computer Engineering (High Distinction)

Melbourne, Australia

Sep. 2010 – Nov. 2010

RMIT University

Master of Electronics Engineering (High Distinction)

Melbourne, Australia

Aug. 2008 – Jul. 2012

Fuzhou University

Bachelor of Science in Applied Physics

Fuzhou, China

Sep 2004 – Jun 2008

PROFESSIONAL SKILLS

- **Programming Languages:** C/C++, Python, F sharp, Bash, HTML, CSS, Javascript, SQL, Objective C, Matlab, VHDL, PLC, Assembly, $\text{\LaTeX}$, LabView
- **OS/Middleware:** Linux, Windows, FreeRTOS, ROS
- **Software Framework/Libraries:** Qt, OpenGL, Protobuf, Boost, Matplotlib, Numpy, Scipy, Pcl, Eigen, Sophus, G2O, Ceres, OpenCV, NodeJs, Pytorch, Tensorflow
- **Software Tools:** CMake, SCon, Conan, Gitlab, Jenkins, Docker, AWS, Git, Confluence, Jira, Office365, QGIS
- **Technical Stack:** Ultrasonic/Radar/Video/Lidar knowledge, Embedded Software Design, Data Analysis and Visualization, modelling and simulation, SLAM, Sensor Fusion, Nonlinear Optimization, Machine learning, Deep learning, Computer Vision